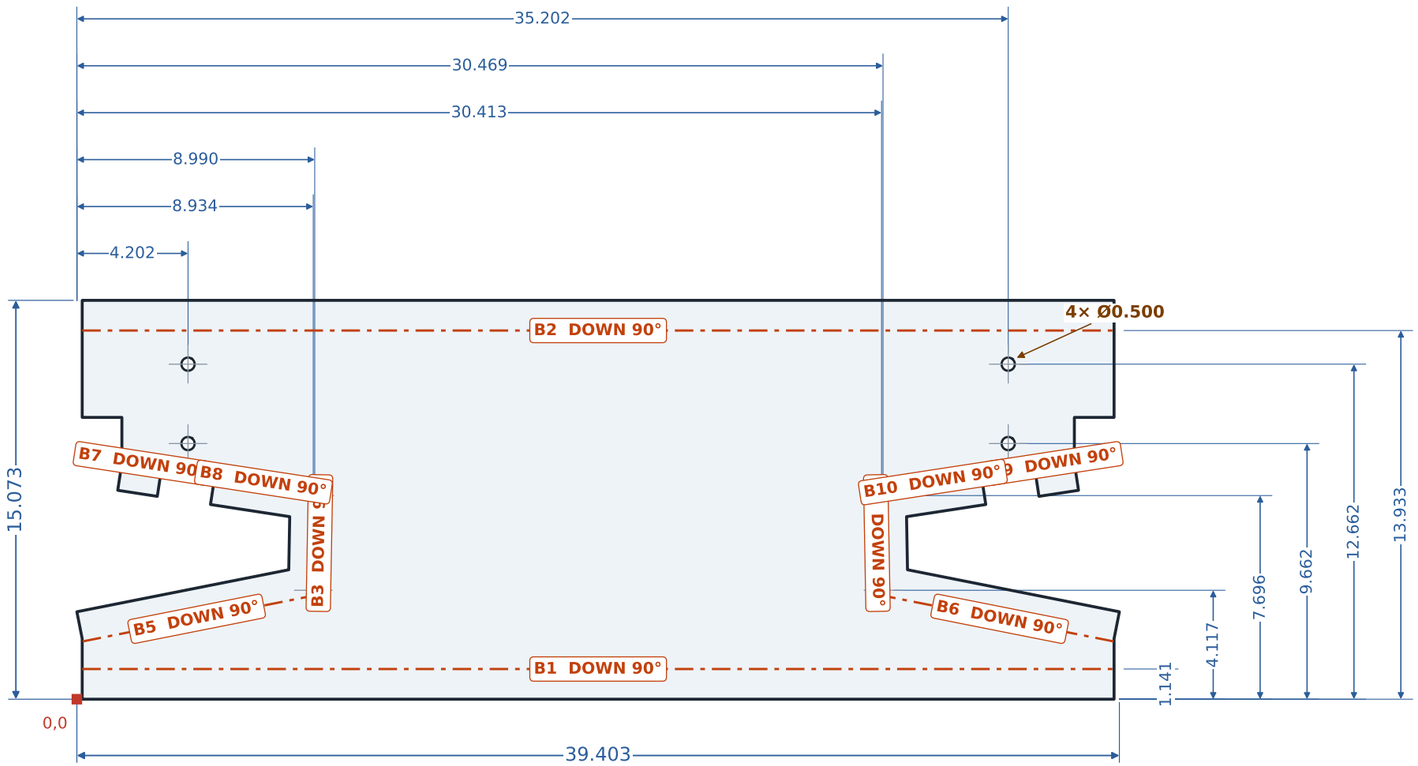
FF-3-1 · FF3-1 top plate — flat pattern, 10 bends 90° down

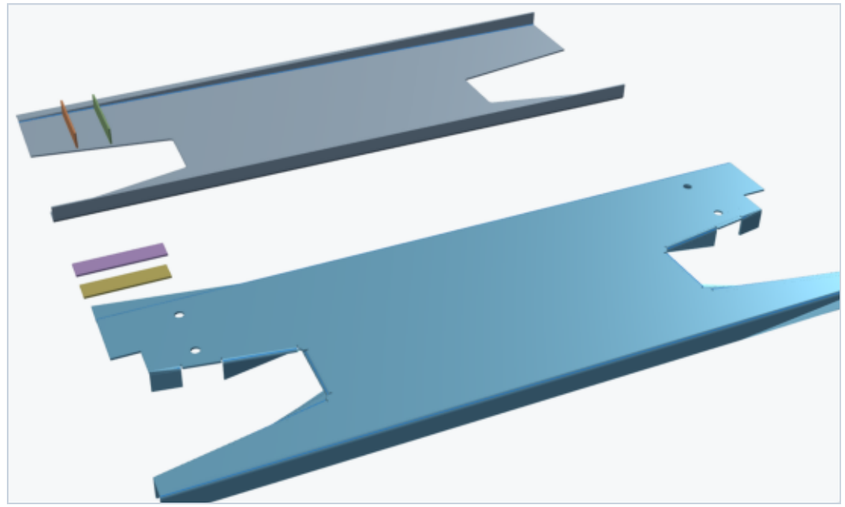
11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 39.403″ × 15.073″ · ≈ 17.70 lb each · QTY 1 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	DOWN (away)	R=t 0.120	3	Y = 1.141
B2	90.0°	DOWN (away)	R=t 0.120	4	Y = 13.933
B3	90.0°	DOWN (away)	R=t 0.120	5	(9.138, 4.256)→(9.197, 7.546)
B4	90.0°	DOWN (away)	R=t 0.120	6	(30.265, 4.256)→(30.206, 7.546)
B5	90.0°	DOWN (away)	R=t 0.120	7	(0.222, 2.185)→(8.833, 3.892)
B6	90.0°	DOWN (away)	R=t 0.120	8	(39.181, 2.185)→(30.570, 3.892)
B7	90.0°	DOWN (away)	R=t 0.120	9	(1.717, 9.022)→(3.212, 8.793)
B8	90.0°	DOWN (away)	R=t 0.120	10	(5.223, 8.486)→(8.878, 7.926)
B9	90.0°	DOWN (away)	R=t 0.120	11	(37.686, 9.022)→(36.191, 8.793)
B10	90.0°	DOWN (away)	R=t 0.120	12	(34.180, 8.486)→(30.526, 7.926)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	cut-out 0.513×0.516	30.326	7.802	laser
H2	Ø0.500	35.202	12.662	laser
H3	Ø0.500	35.202	9.662	laser
H4	cut-out 0.514×0.504	30.387	4.011	laser
H5	cut-out 0.513×0.516	9.077	7.802	laser
H6	cut-out 0.514×0.504	9.016	4.011	laser
H7	Ø0.500	4.202	12.662	laser
H8	Ø0.500	4.202	9.662	laser

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.

* rims 1⅛" bent down on the long sides and around the whole notch; leg ends open; in the 4 inner notch corners the rims are mitred — a gap stays in the corner; 1/16" reliefs at the bend ends; the 4 Ø½ holes are cut right away

FILES

- FF-3-1_cut.dxf (qty 1 / vehicle)
- FF-3-1_ref.dxf (reference: + bend lines, drill-later)

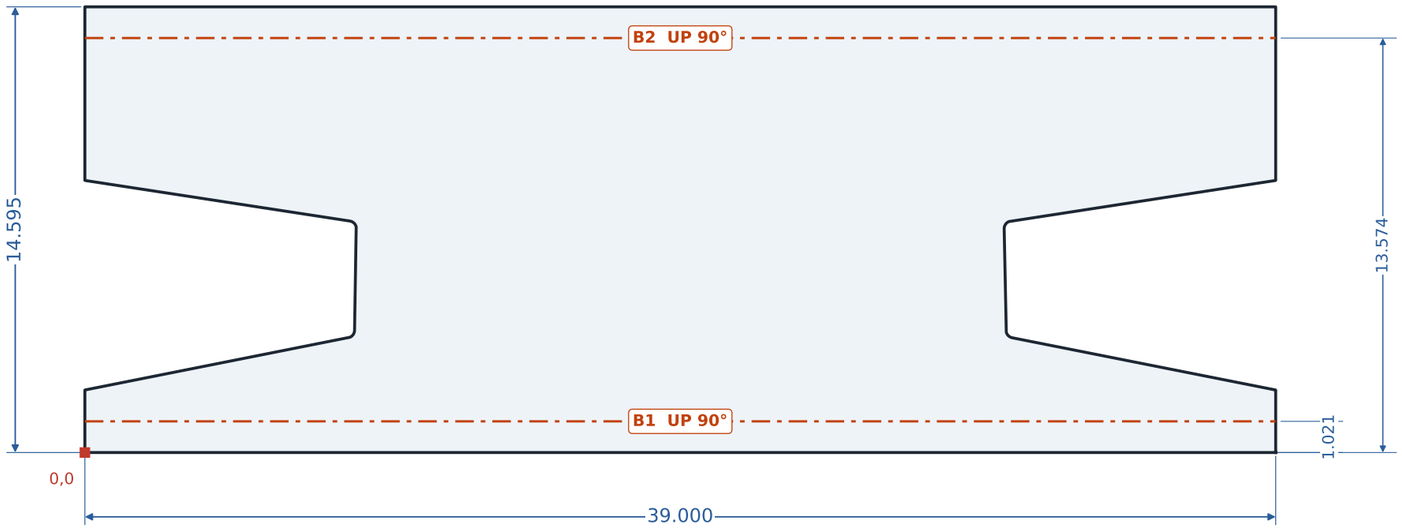
FF-3 Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FF-3-1 · FF3-1 top plate — flat pattern, 10 bends 90° down	**QTY 1 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 1/4**

FF-3-2 · FF3-2 lower plate — flat pattern, 2 bends 90° up

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 39.000″ × 14.595″ · ≈ 16.12 lb each · QTY 1 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196'' (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	Y = 1.021
B2	90.0°	UP (toward)	R=t 0.120	2	Y = 13.574

FILES

- FF-3-2_cut.dxf (qty 1 / vehicle)
- FF-3-2_ref.dxf (reference: + bend lines, drill-later)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.

* rims on the long sides bent up 1" (to the FF3-1 underside), set in by t inside the FF3-1 rims; 1/16" reliefs at the bend ends

FF-3 Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FF-3-2 · FF3-2 lower plate — flat pattern, 2 bends 90° up	**QTY 1 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 2/4**

FF-3-3 · Passage wall 1⅛ × 4.69" (instead of the tube)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 4.687″ × 1.125″ · ≈ 0.18 lb each · QTY 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196'' (3.04 mm)

FILES
- FF-3-3_cut.dxf (qty 2 / vehicle)

FLAT PATTERN (as laser-cut, marked side up)



NOTES
1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.

* stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

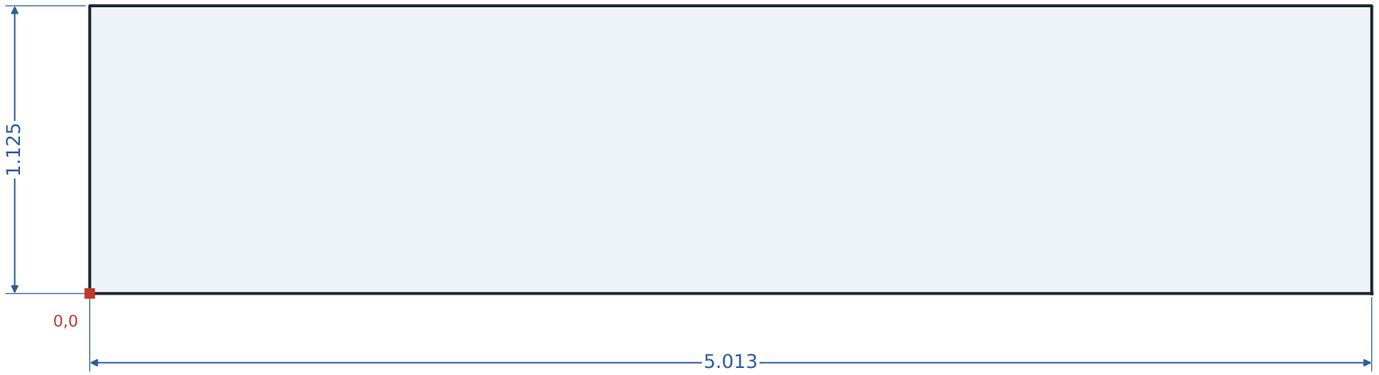
FF-3 Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FF-3-3 · Passage wall 1⅛ × 4.69" (instead of the tube)	**QTY 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 3/4**

FF-3-4 · Passage wall 1⅛ × 5.01" (instead of the tube)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 5.013″ × 1.125″ · ≈ 0.19 lb each · QTY 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196'' (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FILES

- FF-3-4_cut.dxf (qty 2 / vehicle)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.

* stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

FF-3 Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FF-3-4 · Passage wall 1⅛ × 5.01" (instead of the tube)	**QTY 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 4/4**